# SOKOINE UNIVERSITY OF AGRICULTURE

## School of Engineering and Technology

*Department of Agricultural Engineering*

# ABE 721

## Agricultural Systems Modelling and Simulation

**OFFICIAL COURSE SYLLABUS**

**Course Instructor:**

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*Electronics and Precision Agriculture Laboratory (EPAL)*

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# 1. Course Information

## 1.1 Course Details

- **Course Code:** ABE 721
- **Course Title:** Agricultural Systems Modelling and Simulation
- **Course Status:** Core
- **Credit Rating:** 12 Credits
- **Total Hours Spent:** 120 Hours (Lectures, Practicals, and Independent Study)
- **Host Department:** Department of Agricultural Engineering
- **Host School:** School of Engineering and Technology

## 1.2 Course Aim

The primary aim of this course is to impart students with rigorous, postgraduate-level knowledge and computational skills required for developing, simulating, and validating complex agricultural systems using advanced mathematical models and software engineering tools.

## 1.3 Expected Learning Outcomes

Upon the successful completion of this rigorous course, students shall be able to:

1. **Formulate** a real-world agricultural engineering problem and critically select the most appropriate analytical technique for modelling and ultimately solving the problem.
2. **Use** advanced computational simulation software (e.g., MATLAB, Python, Vensim) for the mathematical development and analysis of dynamic agricultural systems.

3. **Design** and successfully run complex, multi-variable simulation models reflecting practical, real-world farm conditions.
4. **Communicate** the results of the mathematical modelling process highly effectively to management, policy-makers, and other non-specialist users through professional scientific writing.

# 2. Assessment and Grading Structure

The assessment strategy for ABE 721 is designed to rigorously evaluate both the student's theoretical comprehension of systems engineering mathematics and their practical ability to synthesize, execute, and document a novel simulation model.

The final grade is composed of three primary pillars:

| Assessment Component | Description | Weight |
|---|---|---|
| **Test 1** | Covers Modules 1, 2, and 3. Open-book policy. | 20% |
| **Test 2** | Covers Modules 4, 5, and 6. Open-book policy. | 20% |
| **Final Term Paper Project** | A comprehensive, peer-review style scientific paper detailing a novel simulation model developed by the student. | 60% |
| | **Total** | **100%** |

## 2.1 Examination Policy (Tests 1 & 2)

- **Open-Book Format:** Due to the heavy reliance on complex calculus, differential equations, and matrix algebra, both Test 1 and Test 2 are strictly **open-book**. Students are permitted (and encouraged) to bring physical textbooks, printed course modules, and handwritten notes into the examination hall.
- **Device Ban:** To maintain academic integrity, the use of **mobile phones, smartwatches, tablets, and laptop computers is strictly prohibited** during the tests. Only standard, non-programmable scientific calculators are permitted.

## 2.2 Continuous Assessment (The 6 Assignments)

Throughout the semester, there will be exactly **6 assignments** (one explicitly assigned at the conclusion of each Module).

- **Bonus / Remedial Function:** These assignments are not factored directly into the base 100% grading scale. Instead, they serve as a critical remedial buffer.
- **Score Improvement:** The cumulative scores of these 6 assignments will be used algorithmically by the instructor to **improve poor test scores** (Test 1 and Test 2) prior to calculating the final grade. Students are heavily advised to complete all assignments with maximum diligence to protect against unexpected poor performance in the written, time-constrained tests.

# 3. Final Term Paper Project (60%)

Because modelling is an applied science, the traditional final written examination has been replaced entirely by a **Term Paper Project** carrying 60% of the total course grade. Students must conceptualize, mathematically formulate, computationally execute, and statistically validate a novel agricultural systems model.

## 3.1 Scientific Paper Format Requirements

The final submission must not read like a homework assignment; it must be formatted as a professional, academic manuscript ready for submission to a peer-reviewed journal (e.g., *Computers and Electronics in Agriculture* or *Agricultural Systems*).

The paper must strictly adhere to the following structural format:

1. **Title Page:** A descriptive, professional title. Student name and registration number.
2. **Abstract (max 300 words):** A concise summary of the problem, the modelling paradigm used, the primary simulation results, and the ultimate conclusion.
3. **1. Introduction:**
    - Definition of the agricultural problem (e.g., pest outbreak, irrigation inefficiency).
    - Literature review of previous models.
    - Explicit statement of the model's objective and system boundaries.
4. **2. Methodology (Model Formulation):**
    - Causal Loop Diagrams (CLDs) or Stock/Flow conceptual diagrams.
    - The explicit mathematical formulation (ODEs, PDEs, or state-space matrices).

- Explanation of the numerical solver used (e.g., RK4, Crank-Nicolson).
- Parameter estimation techniques and probability distributions used.

5. **3. Results and Simulation Analysis:**
    - Professional, high-quality graphs of the simulation outputs.
    - Phase plane analysis or stability proofs (if applicable).
    - Sensitivity analysis (how variations in parameters affect the output).
6. **4. Validation and Discussion:**
    - Comparison of simulated results against real-world data or established literature.
    - Calculation of validation metrics (RMSE, NSE, Percent Bias).
    - Discussion of the model's limitations and boundary flaws.
7. **5. Conclusion:** The final normative or explanatory engineering takeaway.
8. **References:** Strict adherence to APA (7th Edition) or IEEE citation formatting.

*Note: The underlying computer code (Python, MATLAB, or R scripts) must be submitted digitally as an appendix, but the paper itself will be graded solely on the quality of the scientific writing and the mathematical rigor of the methodology.*

# 4. Academic Policies

As postgraduate engineers at the School of Engineering and Technology, students are expected to uphold the absolute highest standards of scientific integrity.

## 4.1 Academic Honesty Clause

Plagiarism, data fabrication, and the falsification of simulation results are considered severe academic offenses.

- **Originality:** All mathematical formulations, causal loop diagrams, and written analysis submitted in the Term Paper and the 6 Assignments must be the student's original work.
- **Citation:** Any mathematical equation, parameter value (e.g., a specific crop coefficient), or block of open-source computer code derived from external literature or existing software repositories must be explicitly and correctly cited.
- **Penalty:** Any submission found to violate this honesty clause will instantly receive a grade of zero (0%) and the student will be reported to the University Disciplinary Committee for further academic sanctions, which may include expulsion from the program.

## 4.2 Artificial Intelligence (AI) Use Policy

The School of Engineering and Technology recognizes that generative Artificial Intelligence (e.g., ChatGPT, GitHub Copilot, Claude) is rapidly becoming an industry-standard tool for software engineers and data scientists. Therefore, the use of AI is **permitted under strict, specific guidelines**:

1. **Permitted AI Uses:**
    - Debugging syntax errors in Python, MATLAB, or R simulation code.
    - Brainstorming initial conceptual structures for Causal Loop Diagrams.

- Understanding complex mathematical derivations or numerical integration algorithms (e.g., asking an AI to explain the Runge-Kutta mechanism).
- Formatting citations or generating LaTeX/TikZ code for data visualization.

2. **Prohibited AI Uses:**
   - Generating the actual prose/text of the final Term Paper. The scientific writing, discussion, and analysis must be generated by the human student.
   - Asking the AI to fully architect and formulate the core differential equations of the novel system. The engineering logic must be your own.
3. **Mandatory Disclosure:** If AI was used in any capacity (including code debugging or data visualization), the student **must include an "AI Acknowledgement Statement"** at the end of their Term Paper, explicitly detailing which tools were used and specifically how they were applied. Failure to disclose AI assistance is considered a violation of the Academic Honesty Clause.

# 5. Course Policies and Assignment Submission

To prepare students for rigorous professional engineering reporting and academic publishing, the following strict guidelines apply to all coursework in ABE 721:

- **LaTeX Requirement:** All assignments, reports, and projects **must** be typeset and submitted as a PDF using LaTeX. **Nothing should be submitted as handwritten.** Handwritten submissions will not be accepted and will automatically receive a grade of zero. Students are highly encouraged to use **Overleaf** (`https://www.overleaf.com`) for an accessible, cloud-based LaTeX editing environment.
- **Programming Submissions:** For assignments requiring computational implementation (e.g., Python, MATLAB, or R), your LaTeX document must explicitly include:
  1. Properly formatted textual explanations of your algorithms.
  2. Embedded, high-quality **screenshots of the code outputs** (including console text results and any generated graphs/plots).
- **Source Code Files:** In addition to the LaTeX-generated PDF document, students must submit the actual, raw executable source code files (e.g., `.py`, `.m`, `.R`) directly to the instructor as part of their final assignment package.

## 5.1 Submission Procedure and Deadlines

✓ **Deadline:** Please submit your work by **one week after the completion of the respective Module**.

✓ **Email Instructions:** Submit all files (the compiled PDF and the raw programming scripts) directly to my email: **kadeghe.fue@sua.ac.tz**.

✓ **CRITICAL:** Do **not** send this work to **kadefue@sua.ac.tz**. Submissions sent to the incorrect address will not be received or graded.

## 5.2 Example of a Systematic Written Answer

To demonstrate the expected standard of work, below is an example of a perfectly formatted assignment response incorporating math, code, and screenshots.

**Example Prompt:** *A water tank is draining according to the ODE $dV/dt = -k\sqrt{V}$, where $k = 0.5$. Initial volume $V_0 = 100$ L. Write a script to calculate the volume at $t = 10$ seconds using the Forward Euler method with $h = 1$ second.*

**Student Answer:**

**1. Mathematical Formulation:**

The Forward Euler algorithm updates the state using the first derivative. We discretize the continuous ODE as follows:

$$V_{n+1} = V_n + h \cdot \left(-k\sqrt{V_n}\right)$$

**2. Computational Implementation (Python):**

```
import numpy as np
import matplotlib.pyplot as plt

V = 100.0  # Initial volume (L)
k = 0.5
h = 1.0    # Time step (s)

# Forward Euler Integration Loop
for t in range(0, 10):
    dV_dt = -k * np.sqrt(V)
    V = V + h * dV_dt

print(f"Volume at t=10s: {V:.2f} Liters")
```

**3. Code Output and Visual Results:**

*[Insert Screenshot of Console Output Here: "Volume at t=10s: 57.64 Liters"]*

*[Insert Screenshot of Generated Plot Figure Here]*

# 6. Required Reading and References

Students are heavily encouraged to consult the following foundational texts, which map directly to the modules explored in this course. To ensure students grasp both the historical mathematical foundations and the modern computational methodologies, the reading list provides the classic syllabus texts alongside their state-of-the-art modern equivalents (published post-2020):

1. **Agricultural Systems and Crop Modelling**
   - *Classic:* Peart, R.M. and Curry, R.B. (1998). *Agricultural Systems Modeling and Simulation.* Marcel Dekker, Inc., New York, NY.
   - *Modern Equivalent (>2020):* Wallach, D., Makowski, D., Jones, J.W., & Brun, F. (2021). *Working with Dynamic Crop Models: Methods, Tools and Examples for Agriculture and Environment* (3rd Edition). Academic Press.
2. **Soft Computing and AI**
   - *Classic:* Tayfur, G. (2014). *Soft Computing in Water Resources Engineering: Artificial Neural Networks, Fuzzy Logic and Genetic Algorithms.* WIT Press.
   - *Modern Equivalent (>2020):* Pande, C. B. et al. (2022). *Artificial Intelligence and Soft Computing for Water Resources Management*. Springer.
3. **Environmental Modelling and System Dynamics**
   - *Classic:* Ford, A. (1999). *Modeling the Environment: An introduction to system dynamics models.* Island Press.
   - *Modern Equivalent (>2020):* Wainwright, J., & Mulligan, M. (Eds.). (2021). *Environmental Modelling: Finding Simplicity in Complexity* (3rd Ed.). John Wiley & Sons.
4. **Biological Systems**
   - *Classic:* Haefner, J.W. (1996). *Modeling biological systems: Principles and applications.* Chapman & Hall.
   - *Modern Equivalent (>2020):* Klipp, E., Liebermeister, W., Wierling, C., & Kowald,

A. (2022). *Systems Biology: A Textbook* (3rd Ed.). Wiley-VCH.

5. **Plant Water Relations**
   - *Classic:* Kramer, P.J. (2012). *Water Relations of Plants.* Academic Press.
   - *Modern Equivalent (>2020):* Kirkham, M. B. (2021). *Principles of Soil and Plant Water Relations* (3rd Ed.). Academic Press.
6. **Irrigation Engineering**
   - *Classic:* Michael, A.M. (2008). *Irrigation: Theory and Practice.* Vikas Publishing House.
   - *Modern Equivalent (>2020):* Waller, P., & Yitayew, M. (2021). *Irrigation and Drainage Engineering* (2nd Ed.). Springer.
7. **Computer Aided Manufacturing (CAM)**
   - *Classic:* Knudra, C.V. (2000). *Numerical Control and Computer Aided Manufacturing.* Tata McGraw Hill.
   - *Modern Equivalent (>2020):* Alavala, C. R. (2022). *CAD/CAM: Concepts and Applications* (2nd Ed.). PHI Learning.
8. **Mechanical Design and CAD**
   - *Classic:* Rammmurty, T. (2001). *Computer Aided Mechanical Design and Analysis.* Tata McGraw Hill.
   - *Modern Equivalent (>2020):* Chang, K. H. (2022). *Product Design Modeling using CAD/CAE: The Computer Aided Engineering Design Series*. Academic Press.
9. **Matrix and Structural Analysis**
   - *Classic:* Mukhopadhyay, M. (2000). *Matrix, Finite Element, Computer and Structural analysis.* Oxford & IBH Publishing.
   - *Modern Equivalent (>2020):* Kattan, P. I. (2023). *MATLAB Guide to Finite Elements: An Interactive Approach*. Springer.
10. **Finite Element Analysis (FEA)**
    - *Classic:* Krishnamooorty, G. (2001). *Finite Element Analysis. Theory and Programming.* Tata McGraw Hill.
    - *Modern Equivalent (>2020):* Reddy, J. N. (2021). *An Introduction to the Finite Element Method* (4th Ed.). McGraw-Hill Education.